EXPERIMENT 4

Implement histogram equalization and compare it with the original image using OpenCV.

Aim

Implement histogram equalization and compare it with the original image using OpenCV.

Algorithm

• Import OpenCV and load the image.

• Convert the image to grayscale.

• Apply histogram equalization.

• Place the original and equalized images side by side.

• Save the comparison image.

Code

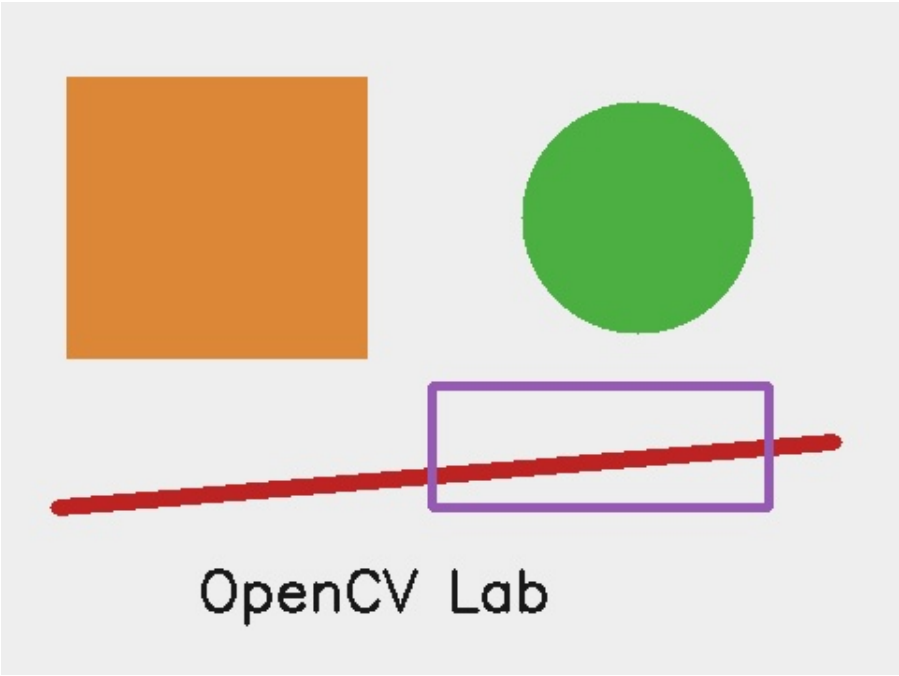
```
import cv2
import numpy as np

img = cv2.imread("images/input.jpg")
if img is None:
    raise FileNotFoundError("images/input.jpg not found")
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
equalized = cv2.equalizeHist(gray)
comparison = np.hstack((gray, equalized))
cv2.imwrite("output_images/experiment04_histogram_equalization.png", comparison)
print("Saved output_images/experiment04_histogram_equalization.png")
```

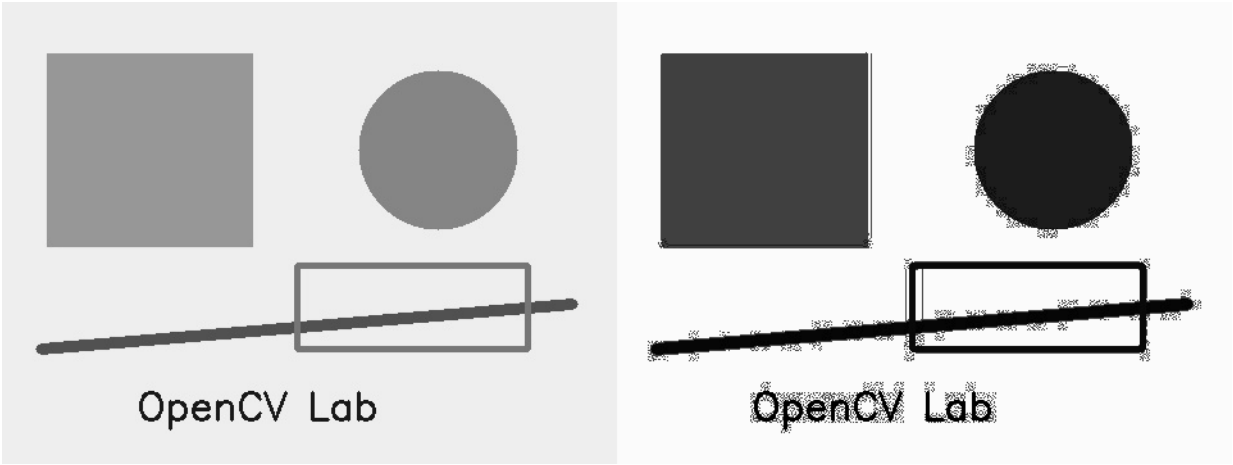
Input Image:

Output Image:

Input Image



Output Image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